TITANIC SURVIVAL PREDICTION USING LOGISTIC REGRESSION

TEAM - 64

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Repo link- https://github.com/2026mcb1515-strgzr/AI_miniproject

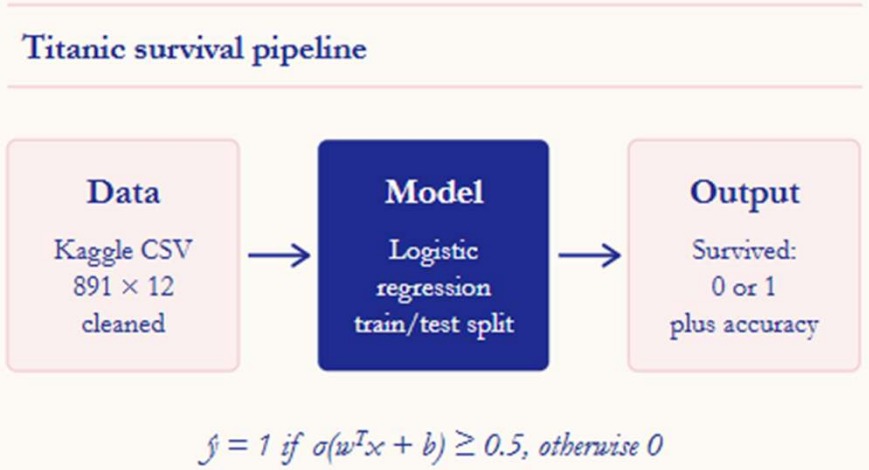
PROBLEM AND DATA

- The problem is to predict the survival of an individual on the titanic with binary classification using logistic regression(0=died, 1=survived).
- Logistic Regression is used because it is a simple interpretable baseline that shows how factors like age, sex and class affected survival on the Titanic.
- The data set is a .csv file obtained from Kaggle - https://www.kaggle.com/datasets/yasserh/titanic-dataset
- It consists of 12 columns: 11 input features plus the target(survival) and has 891 corresponding entries(one per passenger).

APPROACH

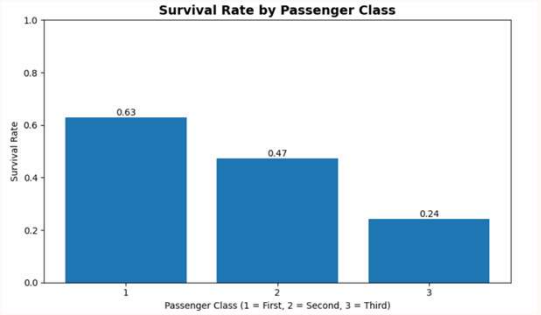
Libraries used:

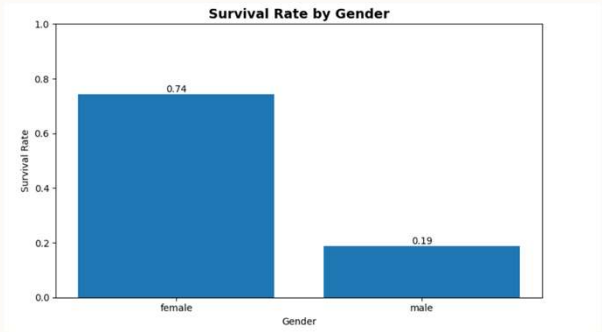
- Pandas: to load the csv file, and to inspect and clean data.
- Numpy: to deal with arrays and calculations.
- Scikit-learn: for the train-test split, logistic regression model and evaluation metrics.
- Matplotlib: for plots and charts
- Seaborn: for statistical plots
- Counter: for counting categories or class labels.



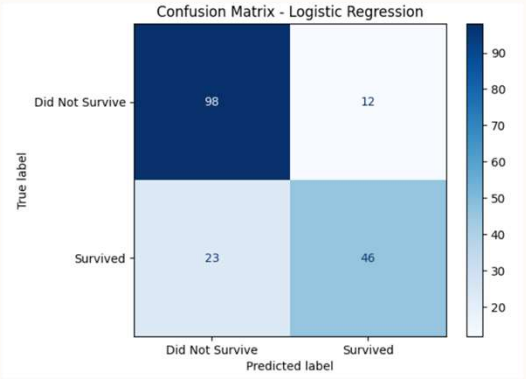
RESULTS

- The model's accuracy is 80.447%.
- Precision and Recall: 79.31% and 66.67%
- F1 Score: 72.44%





Confusion matrix



Coefficients and intercept of the model

	Feature	Coefficient
0	Pclass	-0.928698
1	Sex	1.269598
2	Age	-0.502878
3	SibSp	-0.262672
4	Parch	-0.067158
5	Fare	0.099599
6	Embarked_Q	0.080683
7	Embarked_S	-0.171784

Intercept: -0.657

Demo of the model

```
import pandas as pd

sample_passenger = X_test.iloc[[0]]

prediction = model.predict(sample_passenger)

original_passenger = df.loc[sample_passenger.index]

print("Passenger Details:")
print(original_passenger)

if prediction[0] == 1:
    print("Prediction: Survived")
else:
    print("Prediction: Did Not Survive")
```

```
Passenger Details:
     PassengerId  Survived  Pclass                 Name   Sex   Age  SibSp  \
565          566         0       3  Davies, Mr. Alfred J  male  24.0      2

     Parch     Ticket   Fare Cabin Embarked
565      0  A/4 48871  24.15   NaN        S
Prediction: Did Not Survive
```

LEARNINGS

- What worked:
    - Clean pipeline (data → model → output) made each step easy to check.
    - Fitting StandardScaler on the training data avoided test-set leakage.

- What didn't work:
    - Logistic regression assumes that log odds are a linear combination of the features, which may not be the case.
    - Smaller dataset due to which there are relatively noisier results.
- Who did what:
    - Aryan: Handled the github documentation and loaded the csv file along with required libraries.
    - Akshit: Made the requirements.txt file and preprocessed the data.
    - Goutham: Made the PPT and trained the model.
    - Maitrayee: Made the video and evaluated the model.
- AI disclosure: AI (Claude) was used to generate the pipeline diagram shown on slide 3. It was also used to create roadmap on how to approach the chosen problem, draft certain parts of the preprocessing section, set up the git and python environment and was used in creating an initial README draft.